

The Ion-Transport Landscape of Human Tissues

CFDE Data Visualization Competition 2026

Intended audience: Researchers and clinicians. Physiologists, computational biologists, and anyone who thinks about how tissues actually work.

Narrative

Every cell in your body is moving ions right now. Sodium out, potassium in. Chloride and bicarbonate swapped across the membrane. It never stops, and you never notice it. But it is the reason your liver can clear a drug, your pancreas can neutralize stomach acid, and your gut can pull water out of your food.

The genes that do this work are mostly the same from one organ to the next. What changes is how each tissue lays them out in space. I wanted to see if that difference was real and visible, so I ran one test three ways: the same screen for ion-transporter genes, applied to three different human organs, and then I looked at where those genes were switched on.

The data come from SenNet and HuBMAP, two NIH Common Fund programs, which I reached through the CFDE ecosystem. I used Xenium spatial transcriptomics from human liver, pancreas, and large intestine. That is more than 375,000 cells in total, each one drawn at the exact spot it occupies in the tissue. To pick the transport genes without cherry-picking, I queried UniProt and MyGene for ion-transport Gene Ontology terms and used the same rules on every dataset. Then I gave each cell one number: how strongly it expresses its transport program. I call it the Ion-transporter Expression Aggregate, or IEA.

I have to be straight about what that number is. It comes from messenger RNA. It tells you a cell has switched on its transport genes. It does not tell you ions are actually flowing. A gene being on is not the same as a channel doing its job. So everywhere I show "flow," I mean the direction in which expression rises across the tissue. It is a gradient of gene activity, not a current of ions, and I label it that way every time.

What comes out of this is hard to look away from. The liver sorts itself into zones. The pancreas draws its ductal tree against a background of acinar cells. The intestine builds

crypts, and you can actually see the rings, with absorptive cells up top and secretory cells at the base. Same set of genes, three different answers. Each one matches what anatomists have described for over a century.

I added a second method to check my own work. Xenium sees single cells but reads only a handful of transport genes. GeoMx reads the whole transcriptome, and the same screen finds 46 ion genes instead of a dozen, including proton pumps and bicarbonate transporters that targeted imaging never sees. It confirms the liver's periportal-to-pericentral gradient on its own ($p = 0.003$, holding up across all four regions I sampled). Neither method is complete by itself. Put together, they say the same thing.

Transport is not a flat, even hum across a tissue. It is organized into ducts, crypts, and zones. And that organization is written into the transcriptome, one cell at a time, all over the body.

Data: SenNet and HuBMAP (NIH Common Fund) via CFDE. IEA reflects mRNA expression, a proxy for transport capacity, not measured ion flux. Cell groups are defined by canonical transporter-marker expression, not reference-validated annotation.